Experience Day - Full Stack Developer - HeatOS

Building a Multi-Channel Lead Intake Funnel

Scenario:

Vamo has an established semi-automated sales funnel that accepts leads from multiple sources such as our own website or through lead partners.

Your Mission:

Your task is to prototype a “Lead Intake Funnel” consisting of a backend service and a frontend client which forward Leads to a CRM[1].

[1] We do know that you have incomplete information, so please make assumptions if you need to and highlight where you made the assumptions and why.

Your Task:

Your final deliverable will be a prototype demonstration and a discussion outline for our two-hour call. Focus on the analysis, design, and implementation. A full documentation is not required, but encouraged if you want to show your skills in that area.

🛠️ **Choice of Technology**
Vamo runs on a stack of Vue.js, Nest.js and MongoDB hosted on Heroku via dockerized Containers, all written in TypeScript. We strongly recommend you to use the technologies mentioned above for this task.
However, we acknowledge that other technologies are a great fit as well and therefore give you the option to use other tools/technologies. If you deviate from the Vamo Tech-Stack we’d be interested to understand your reasons behind it.

🤖 Use of AI
You are free to use any AI tools or Agents for this task. However, we expect full disclosure which parts of the task were done with the help of AI.

Part 1: Backend Implementation

You are required to implement a RESTful backend service that accepts lead data and persists it.

- **API Specification**: Please refer to the official API definition here: **Lead Creation v1.2.0 (Notion)**
- **Requirements**:
  - Implement an endpoint as described in the documentation above.
  - Ensure the payload validation matches the rules specified in the docs
    - For brevity you can skip the `selling` stage and focus on `qualification` and `discovery`.
  - Forward the received Leads to CRM (preferrably Salesforce, however you can use any CRM you like or simply store them in a Database)
  - Bonus: Extend the API to allow uploading of pictures

Part 2: Frontend Implementation

Create a simple frontend application that allows a user to input their details and submit them to your backend.

- **Requirements**:
  - Create a form that includes all fields required by the Backend API (e.g., Name, Email, etc.). You can copy the form on https://vamo-energy.com/rechner if you like
  - Implement client-side validation
  - **Bonus**: Accept pictures uploads
  - **Bonus 2**: Make the form offline capable to take pictures where there is no internet connection and upload later on submitting the form

Part 3: Infrastructure & Scalability Strategy

Please answer the following questions regarding the infrastructure required to run this system.

Explain your choice of technology stack (hosting, database, etc.) and the infrastructure setup you would use for the following two stages of growth:

1. **Stage A: 5,000 leads per month**
   - *Focus:* Speed to market, low cost, simplicity.
   - *Question:* Where would you host the frontend and backend? What database tier would you use? How much would this roughly cost?
2. **Stage B: 500,000 leads per month**
   - *Focus:* Reliability, observability, stability.
   - *Question:* What changes (if any) would you make to the infrastructure? How would you handle monitoring, logging, and backups?

Deliverables for the Call:

> 👸 **Content is King**
> While we appreciate a good looking document, we appreciate a sharp train of thought more. So when pressed for time, please focus on your content and skip any design work.

Primarily we want to see your code and demo, if you want to communicate your Ideas via a short presentation, feel free to do so, but it is not required.

Presentation Guidelines:

The two-hour call will be a collaborative discussion. The first 30-45 minutes will be dedicated to your work and a Q&A session. The remaining time will be a deeper dive into your thought process, and we'll work together on a hypothetical new challenge to see how you think on your feet.

We are not looking for a perfect solution, but rather a demonstration of your ability to think holistically about process, people, and technology in a fast-paced environment.